



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Homework — Series and Parallel Circuits

Every question here can be answered with $V = IR$ and the rules you measured in class. Show every step: an answer with no working earns almost nothing in the exam, and I cannot see where you went wrong.

- 1) **State** the two rules for a series circuit — one about current, one about potential difference — and the equation for the total resistance of two resistors in series. [3]

- 2) **Calculate** the total resistance, and then the current, in a series circuit where a $4\ \Omega$ resistor and a $2\ \Omega$ resistor are connected to a $12\ \text{V}$ supply. [3]

- 3) **Calculate** the potential difference across *each* resistor in the circuit from question 2, and show that your two answers are consistent with the supply. [3]

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4) **Calculate** the current in each branch, and the total current from the supply, when a $3\ \Omega$ resistor and a $6\ \Omega$ resistor are connected *in parallel* across a 6 V supply. [4]

Parallel: Each branch gets the *full* supply pd — so use $V = IR$ on one branch at a time. You are *not* being asked for a total resistance: Combined Science never asks you to calculate that for parallel resistors.

5) **Explain** why adding a resistor *in series* increases the total resistance of a circuit, but adding a resistor *in parallel* decreases it. No calculation is needed. [4]

Think: How many routes can the charge take in each case?

6) **Explain** why the lights in a house are wired in parallel rather than in series. Refer to potential difference in your answer. [4]

Total: 21 marks. Due Friday 11 September — bring it to the lesson. If you are stuck, start from $V = IR$ and the rules table on the circuits handout.